INCH POUND

MIL-C-3098/103C
5 May 1993
SUPERSEDING
MIL-C-3098/103B
12 January 1987

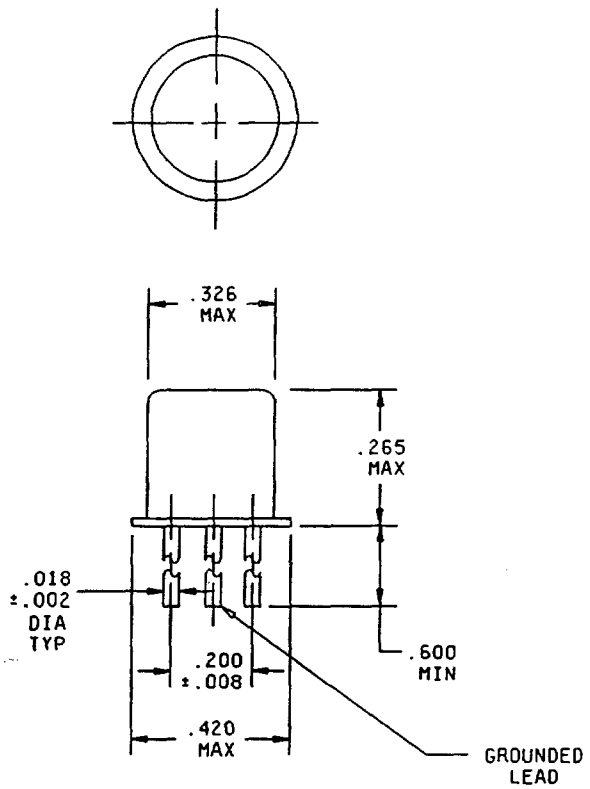
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR101/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 7 to 20 MHz; fundamental; noncontrolled; antiresonance; cold weld sealed.



Inches	mm
.001	.03
.002	.05
.005	.13
.0185	.470
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR101/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-35/U (MIL-H-10056/21).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 7 to 20 MHz, inclusive.

Frequency tolerances: ±50 ppm.

Equivalent resistance: 7 to 10 MHz: 30 ohms.

10+ to 20 MHz: 25 ohms.

Antiresonance, load capacitance: 30 ±0.5 pF.

Mode of oscillation: Fundamental.

Operating temperature range: -55°C to +105°C, inclusive.

Drive level: 5.0 ±1.0 mW.

Capacitance, shunt: 7 pF, maximum.

Shock (specified pulse):

Permitted frequency change: ±5 ppm.

Permitted equivalent resistance change: ±10 percent.

Thermal shock:

Permitted frequency change: ±5 ppm.

Permitted equivalent resistance change: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition B.

Permitted frequency change: ±5 ppm.

Permitted equivalent resistance change: ±10 percent.

Aging:

Permitted frequency change: 25 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0678-01)

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